JMnet: Joint Metric Neural Network for Hyperspectral Unmixing

Anyou Min, Ziyang Guo, Hong Li, *Member, IEEE*, and Jiangtao Peng

***Abstract*— Hyperspectral unmixing is a significant task in remote sensing image analysis. Existing learning-based methods for hyperspectral unmixing generally are in the form of an autoencoder and take geometric distances, such as spectral angle distance (SAD) as loss functions. These methods ignored the distribution similarity between the observation and the reconstruction, which might help improve the unmixing performance. Besides, the autoencoder is trained by directly comparing the difference between the observation and the reconstruction, and the difference between their features has been neglected. Based on the above considerations, we propose a joint metric neural network for hyperspectral unmixing, by introducing the *Wasserstein distance* and *feature matching* as regularization terms and SAD as the underlying loss. The proposed neural network consists of two parts: an autoencoder is used for endmember extraction and abundance estimation, while a discriminator is used to compute the Wasserstein distance. The Wasserstein distance can stably provide useful gradient information that promotes the autoencoder to reach a solution with better unmixing performance. The feature matching is adapted to an intermediate layer of the discriminator for enforcing the features of the observation and the reconstruction to be equal, which can lead to further improvement of the unmixing performance. The model analysis and the regularization parameter analysis are conducted to demonstrate the effectiveness of our method. Experimental results on four real-world hyperspectral data sets show that our method outperforms the state-of-the-art methods, especially in terms of abundance estimation.**

***Index Terms*— Feature matching, hyperspectral unmixing, neural network, Wasserstein distance.**

I. Introduction

HYPERSPECTRAL images (HSIs) contain hundreds of continuous and narrow spectral bands beyond the visible spectrum, which have been applied to many fields, such as classification, target detection, and environmental monitoring [1]–[6]. Because of their inherent characteristics, such as low spatial resolution, microscopic material mixing, and multiple scattering, each pixel of the HSIs is typically a mixture of several materials, which will bring difficulties to further research on the HSIs. Therefore, we need to conduct the hyperspectral unmixing (HSU) task that decomposes the mixed pixel into a collection of constitute materials (endmembers) and their corresponding fractions (abundances).

Past decades have developed numerous unmixing algorithms based on linear mixing models (LMMs) and non-LMMs (NLMMs) [7]. The LMMs consider a situation where the mixing takes place on a macroscopic scale, and there is only one material interacting with the incident light, while the NLMMs [8] seek to simulate physical interactions of the light scattered by two or more materials. However, the unmixing approaches based on NLMM have many challenges, such as requiring complex metrics, high computational cost, and prior knowledge of the scene. For simplicity, most unmixing algorithms adopt the LMM assumption, which provides fine approximations under the light scattering mechanisms in a number of real-world scenarios [9].

Traditional linear unmixing methods can be categorized into pure-pixel-based and nonpure-pixel-based [10]–[18]. Pure-pixel-based approaches consider the distribution of hyperspectral pixels as a simplex and regard the vertices of the simplex as endmembers. Classical methods include pixel purity index (PPI) [10], N-FINDR [11], and the vertex component analysis (VCA) [12]. However, the pure-pixel assumption is not always available in practice when the spatial resolution of HSIs is relatively low. Meanwhile, one needs to apply approaches such as FCLS [19] and MLM [20] to calculate the abundances after endmember extraction; what is more, the performance of obtained abundances highly relies on the quality of the extracted endmember signatures.

In order to address these issues, non-pure-pixel-based methods are delivered, including the independent component analysis (ICA), the nonnegative matrix factorization (NMF), and so on. Among them, NMF has played an important role in the HSU task in virtue of its powerful ability for unsupervised unmixing. Qian *et al*. [13] proposed an $L_{1/2}$-NMF method through imposing $L_{1/2}$-norm on the abundance matrix to enhance its sparsity. A robust collaborative NMF (R-CoNMF) is developed by Li *et al*. [14] that can estimate the number of endmember via adding a collaborative regularization. Others [15]–[17] utilized manifold learning to explore the connection between the observation matrix and the abundance matrix.

Recent years have witnessed growing attention on learning-based approaches to perform HSU task, which is in the form of an autoencoder [21]–[28]. The encoder maps the input spectra into abundance fractions; meanwhile, the decoder

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uses a linear mapping to reconstruct the input spectra, where the linear mapping is viewed as endmembers. This type of method mainly focuses on the modifications of the network architecture and loss function. Ozkan *et al*. [24] presented a two-staged autoencoder network by combining SAD with multiple penalty terms as loss function to improve the unmixing performance. Su *et al*. [25] integrated stacked autoencoders and a variational autoencoder to perform hyperspectral unmixing with mean square error (MSE) objective. Dou *et al*. [26] developed an autoencoder merging an orthogonal sparse prior with a hyper-Laplacian loss as the final objective. Elkholy *et al*. [27] constructed a 1-D deep convolutional autoencoder to conduct hyperspectral unmixing, while Palsson *et al*. [28] built a 2-D version, and they both use SAD as the objective. Even these works have achieved great progress in the HSU task; there remains a need for further improvement.

First, the above learning-based methods, which take geometric distances as loss functions, ignore the distribution similarity between the observation and the reconstruction. Second, the autoencoder is trained by directly comparing the difference between the observation and the reconstruction; the difference between their features has been neglected. To cope with these two issues, we propose a joint metric neural network for hyperspectral unmixing in this article.

For the first problem, we employ the Wasserstein distance as the distribution similarity metric due to its powerful transferring ability between distributions [29]. The original Wasserstein distance is highly intractable, but, due to the Wasserstein generative adversarial network [30], we can use an additional network called discriminator to compute it implicitly. The detailed implementation is shown in Section III. For the second issue, we introduce a technique called feature matching to solve it. Motivated by deep metric learning [31], we not only pursuit high similarity between the observation $\boldsymbol{x}_o$ and the reconstruction $\boldsymbol{x}_r$ but also hope that their corresponding latent characteristics are close to each other. Specifically, let $f(\boldsymbol{x})$ represent the activations of an intermediate layer in the discriminator, and the feature matching is to compare the feature difference between $\boldsymbol{x}_o$ and $\boldsymbol{x}_r$ by mean square error $(1/L)\|f(\boldsymbol{x}_o)-f(\boldsymbol{x}_r)\|_2^2$, where L is the length of vector $f(\boldsymbol{x}_o)$. It is worth noting that SAD is the underlying loss in our model, and the Wasserstein distance and feature matching are regarded as regularization terms.

In summary, the contributions of this article are threefold.

1) We attempt to combine the geometric distance SAD with the distribution similarity metric Wasserstein distance as the loss function to perform hyperspectral unmixing. The Wasserstein distance can stably provide useful gradient information that promotes our model to reach a solution with better unmixing performance. We conduct model analysis and regularization parameter analysis to illustrate this viewpoint.

2) We proposed a network architecture that consists of two parts: an autoencoder and a discriminator. The former is used for endmember extraction and abundance estimation; the latter is utilized to implement the calculation of the Wasserstein distance and feature matching. The regularization parameter analysis shows that the feature matching indeed helps improve our model's performance.

3) Experimental results on four real hyperspectral data sets show that our method obtains better unmixing performance compared with the state-of-the-art methods, especially in terms of abundance estimation.

The remaining of this article is organized as follows. Section II revisits the LMM and introduces some basic operations and metrics in the neural network. Section III elaborates on the proposed method and model analysis. Section IV presents the experimental results and discussions for four real-world data sets. Finally, the conclusion and future work will be drawn in Section V.

II. Preliminary Knowledge

A. Linear Mixing Model

Let $\boldsymbol{X} = [\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_P] \in \mathbb{R}^{B\times P}$ represent an input HSI with B spectral bands and P pixels; $\boldsymbol{x}_p$ stands for the pth pixel where $p = 1,\ldots,P$. R denotes the number of endmember, and $\boldsymbol{e}_r$ means the rth endmember, where $r = 1,\ldots,R$. Each column of matrix $\boldsymbol{E} \in \mathbb{R}^{B\times R}$ indicates an endmember. The abundance vector for $\boldsymbol{x}_p$ will be denoted by $\boldsymbol{m}_p = [m_{1,p}, \ldots, m_{R,p}]^T$, and $\boldsymbol{M} \in \mathbb{R}^{R\times P}$ represents abundance matrix for all the pixels.

The LMM [7] can be formulated as follows:

$$\boldsymbol{x}_p = \sum_{r=1}^{R} m_{r,p}\boldsymbol{e}_r + \boldsymbol{n}_p = \boldsymbol{E}\boldsymbol{m}_p + \boldsymbol{n}_p \tag{1}$$

where $\boldsymbol{n}_p$ represents the noise and assuming that

$$m_{r,p} \geq 0 \quad \forall r, \forall p \tag{2a}$$

$$\sum_{r=1}^{R} m_{r,p} = 1 \quad \forall p. \tag{2b}$$

Then, (1) can be written as

$$\boldsymbol{X} = \boldsymbol{E}\boldsymbol{M} + \boldsymbol{N}. \tag{3}$$

B. Operations in a Neural Network

The rectified linear unit (ReLU) and leaky-ReLU are the most popular activation functions in deep neural networks, though simple but effective. Their definitions are as follows:

$$\varphi(x) = \max(0, x) \tag{4a}$$

$$\phi(x) = \max(\mu x, x) \tag{4b}$$

where φ and ϕ correspond to ReLU and Leaky-ReLU, respectively. μ is a small constant, and $x \in \mathbb{R}$.

Batch normalization (BN) is a learning method used to make deep neural networks faster and more stable through normalization of the input layer by recentering and rescaling [32]. Given a set of vectors $\{\boldsymbol{v}_i\}_{i=1}^m$, the BN can be expressed as

$$\mathrm{BN}(\boldsymbol{v}_i) = \gamma\,\hat{\boldsymbol{v}}_i + \beta \tag{5}$$

where

$$\hat{\boldsymbol{v}}_i = \frac{\boldsymbol{v}_i - \mu_{\mathcal{B}}}{\sqrt{\sigma_{\mathcal{B}}^2 + \epsilon}}$$

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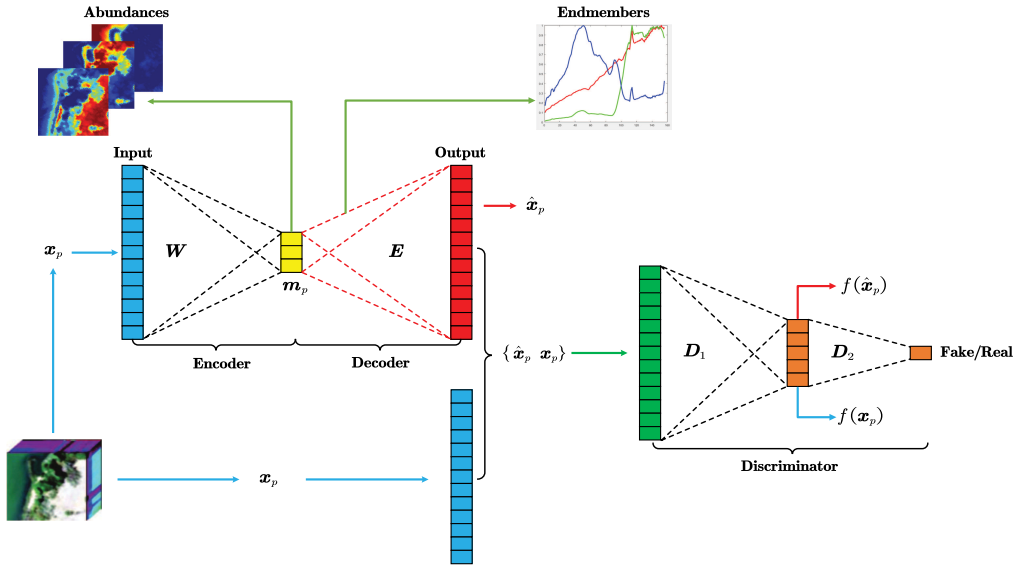


Fig. 1. Pipeline of JMnet. The autoencoder maps the input $\boldsymbol{x}_p$ into abundance vector $\boldsymbol{m}_p$ using a dense layer through BN operation, the ReLU activation function, and the sum-to-one constraint, while the reconstruction process transforms $\boldsymbol{m}_p$ into $\hat{\boldsymbol{x}}_p$ by a linear mapping. Feed $\boldsymbol{x}_p$ and $\hat{\boldsymbol{x}}_p$ into the discriminator alternately. The intermediate layer features $f(\boldsymbol{x}_p)$ and $f(\hat{\boldsymbol{x}}_p)$ are obtained after a dense layer with Leaky-ReLU activation function, and they will be used for feature matching. The last layer of discriminator is linear and its output is a scalar. Intuitively, $\hat{\boldsymbol{x}}_p$ could be seen as a fake sample, while $\boldsymbol{x}_p$ is real. The discriminator is used to compute the Wasserstein distance between $\boldsymbol{x}_p$ and $\hat{\boldsymbol{x}}_p$.

$$\mu_{\mathcal{B}} = \frac{1}{m}\sum_{i=1}^{m} \boldsymbol{v}_i$$
$$\sigma_{\mathcal{B}}^2 = \frac{1}{m}\sum_{i=1}^{m} (\boldsymbol{v}_i - \mu_{\mathcal{B}})^2$$

γ and β are parameters that need to be learned, and ϵ is a small constant. m can be seen as a batch size.

C. Metric

1) *Spectral Angle Distance (SAD):*

$$\mathrm{SAD}(\boldsymbol{a}, \hat{\boldsymbol{a}}) = \arccos\left(\frac{\boldsymbol{a}\cdot\hat{\boldsymbol{a}}}{\|\boldsymbol{a}\|_2\|\hat{\boldsymbol{a}}\|_2}\right). \tag{6}$$

2) *Kullback–Leibler (KL) Divergence:*

$$\mathrm{KL}(\mathbb{P}_r\|\mathbb{P}_g) = \int_{\mathcal{X}} P_r(\boldsymbol{x})\log\left(\frac{P_r(\boldsymbol{x})}{P_g(\boldsymbol{x})}\right)d\boldsymbol{x}. \tag{7}$$

3) *Jensen–Shannon (JS) Divergence:*

$$\mathrm{JS}(\mathbb{P}_r, \mathbb{P}_g) = \mathrm{KL}\left(\mathbb{P}_r\middle\|\frac{\mathbb{P}_r+\mathbb{P}_g}{2}\right) + \mathrm{KL}\left(\mathbb{P}_g\middle\|\frac{\mathbb{P}_r+\mathbb{P}_g}{2}\right). \tag{8}$$

4) *Wasserstein Distance:*

$$\mathrm{W}(\mathbb{P}_r, \mathbb{P}_g) = \inf_{\gamma\in\Pi(\mathbb{P}_r,\mathbb{P}_g)} \mathbb{E}_{(\boldsymbol{x},\boldsymbol{y})\sim\gamma}[\|\boldsymbol{x}-\boldsymbol{y}\|] \tag{9}$$

where $\boldsymbol{a}$, $\hat{\boldsymbol{a}}$, $\boldsymbol{x}$, and $\boldsymbol{y}$ are vectors, and $\cdot$ denotes the vector inner product. $\mathbb{P}_r$ and $\mathbb{P}_g$ are probability measures defined on a compact set $\mathcal{X}$; their densities correspond to P_r and P_g. $\Pi(\mathbb{P}_r, \mathbb{P}_g)$ represents the set of all joint distributions $\gamma(\boldsymbol{x}, \boldsymbol{y})$ whose marginals are $\mathbb{P}_r$ and $\mathbb{P}_g$, respectively, and $\mathbb{E}$ represents the expectation.

TABLE I

ARCHITECTURE OF THE PROPOSED NETWORK. THE NOTATION R INDICATES THE NUMBER OF ENDMEMBER, AND B STANDS FOR THE SIZE OF INPUT SPECTRA

	Layers	Operation	units
Autoencoder	Input layer	-	B
	Hidden layer	BN + φ + STO	R
	Output layer	-	B
Discriminator	Input layer	-	B
	Hidden layer	ϕ	256
	Output layer	-	1

III. PROPOSED METHOD

In this section, we first construct an *autoencoder* for endmember extraction and abundance estimation and apply the batch normalization to speed up training. Then, an additional network called *discriminator* is used to implicitly learn the Wasserstein distance between the original data and the reconstruction. Finally, the total loss is described and followed by model analysis. For ease of understanding, Fig. 1 shows the flow of the proposed method, and a detailed layer-by-layer definition is shown in Table I.

A. Autoencoder

The autoencoder is a three-layer feedforward neural network that is used to learn identity mapping. It contains two parts: the encoder learns a hidden low-dimensional representation of the input, while the decoder maps the hidden representation into an approximation of the input [33], [34]. In the same way, we can model the unmixing process as seeking for a low-dimensional representation of the spectral data, i.e., an abundance map that is used to reconstruct the data. So far,

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the bridge between spectral unmixing and autoencoder has been built. Therefore, we could perform spectral unmixing using autoencoder in practice.

Formally, given a group spectral vectors $\{\boldsymbol{x}_p\}_{p=1}^m \subset \boldsymbol{X}$, the encoding process maps $\boldsymbol{x}_p$ into abundance vector $\boldsymbol{m}_p$, which can be formulated as follows:

$$z_p = \varphi(\mathrm{BN}(\boldsymbol{W}\boldsymbol{x}_p)) \tag{10}$$

$$\boldsymbol{m}_p = \mathrm{STO}(z_p) \tag{11}$$

where $\boldsymbol{W}$ stands for encoder weight matrix connecting the input layer with the hidden layer. φ stands for ReLU in order to satisfy nonnegativity (2a). STO represents the sum-to-one constrain (2b) that could be satisfied through the ℓ_1-norm or the softmax function. That is

$$\boldsymbol{m}_p = \ell_1 - \mathrm{norm}(z_p) = \frac{z_p}{\left(\|z_p\|_1 + \epsilon\right)} \tag{12}$$

$$\boldsymbol{m}_p = \mathrm{softmax}(z_p) = \frac{e^{z_p}}{\|e^{z_p}\|_1} \tag{13}$$

where ϵ is a very small constant to avoid zero-division in the formulation. Since what we considered is a linear model, the reconstruction process should be a linear transformation

$$\hat{\boldsymbol{x}}_p = \boldsymbol{E}\boldsymbol{m}_p \tag{14}$$

where $\boldsymbol{E}$ is the decoder weight matrix (estimated endmembers) connecting the hidden layer with the output layer and $\hat{\boldsymbol{x}}_p$ stands for the reconstructed spectral vector. The whole process can be formulated simply as

$$\hat{\boldsymbol{x}}_p = \mathcal{A}_\theta(\boldsymbol{x}_p) \tag{15}$$

where θ stands for all traning parameters in the autoencoder, i.e., $\theta = [\boldsymbol{W}, \boldsymbol{E}]$.

Due to the scale invariance of SAD, some learning-based researches utilize it as the objective function in hyperspectral unmixing [23], [24], [28]. For this reason, we also apply the SAD as our underlying loss function in this article, that is,

$$\mathcal{L}_{\mathrm{sad}} = \mathbb{E}[\mathrm{SAD}(\boldsymbol{x}_p, \hat{\boldsymbol{x}}_p)] \tag{16}$$

where $\mathbb{E}$ represents the expectation.

B. Discriminator

The Wasserstein distance was first proposed by Gaspard Monge in the optimal transport problem, which seeks an optimal transport cost between the two distributions [29]. Nakhostin *et al*. [35] first attempted to use a discrete version of the Wasserstein distance for supervised unmixing. In contrast, we focus on its continuous version. Since the Wasserstein distance is able to deal with the problem when the supports of two distributions are disjoint, we aim to measure the distribution similarity between $\{\boldsymbol{x}_p\}_{p=1}^m$ and $\{\hat{\boldsymbol{x}}_p\}_{p=1}^m$ by the Wasserstein distance. However, the infimum in (9) is highly intractable. Fortunately, the Kantorovich–Rubinstein duality [29] tells us that

$$\mathrm{W}(\mathbb{P}_r, \mathbb{P}_g) = \sup_{\|h\|_L \le 1} \mathbb{E}_{\boldsymbol{x}\sim\mathbb{P}_r}[h(\boldsymbol{x})] - \mathbb{E}_{\boldsymbol{x}\sim\mathbb{P}_g}[h(\boldsymbol{x})] \tag{17}$$

where the supremum is on all the 1-Lipschitz functions $h: \mathbb{R}^n \to \mathbb{R}$. The h satisfies that $|h(\boldsymbol{x}_1) - h(\boldsymbol{x}_2)| \le \|\boldsymbol{x}_1 - \boldsymbol{x}_2\|$ for $\forall \boldsymbol{x}_1, \boldsymbol{x}_2 \in \mathbb{R}^n$. Intuitively, the 1-Lipschitz imposes an upper bound on the value of $\nabla_{\boldsymbol{x}} h(\boldsymbol{x})$ for $\forall \boldsymbol{x} \in \mathbb{R}^n$.

Now comes the question that how to find the function h solving the problem in (17). To roughly approximate this, a neural network parameterized with weights is utilized, as we would do with a typical generative adversarial network (GAN) [36]. The original GAN lacks training stability since the KL-divergence is discontinuous. We apply the Wasserstein GAN (WGAN) into our unmixing model since the Wasserstein distance has better smoothness than KL-divergence [30]. As a result, our model parameters can reach a solution with better unmixing performance without a special initialization. Specifically, the Wasserstein distance between $\{\boldsymbol{x}_p\}_{p=1}^m$ and $\{\hat{\boldsymbol{x}}_p\}_{p=1}^m$ can be approximated as the following formula:

$$\mathcal{L}_{\mathrm{wd}} = \max_{\omega}[\mathbb{E}(\mathcal{D}_\omega(\boldsymbol{x}_p)) - \mathbb{E}(\mathcal{D}_\omega(\hat{\boldsymbol{x}}_p)] \tag{18}$$

where $\mathcal{D}_\omega$ denotes a neural network so-called discriminator parameterized with ω. Generally, there exists a clipping operation on the weight ω, so as to preserve 1-Lipschitz. However, we practically found that better performance can be obtained without this operation in our unmixing model.

Here, we construct a three-layer discriminator, and its weights are $\boldsymbol{D}_1, \boldsymbol{D}_2$. The feature of the intermediate layer is denoted as

$$f(\boldsymbol{x}) = \phi(\boldsymbol{D}_1\boldsymbol{x}) \tag{19}$$

where $\boldsymbol{x} \in \{\boldsymbol{x}_p, \hat{\boldsymbol{x}}_p\}$, and ϕ represents the Leaky-ReLu function with $\mu = 0.02$ in our model. The WGAN removes the nonlinear activation function at the last layer compared with original GAN, that is,

$$d_{\mathrm{out}} = \boldsymbol{D}_2 f(\boldsymbol{x}) \tag{20}$$

we can simplify the discriminator as

$$d_{\mathrm{out}} = \mathcal{D}_\omega(\boldsymbol{x}) \tag{21}$$

where d_{out} is a scalar and $\omega = [\boldsymbol{D}_1, \boldsymbol{D}_2]$.

In addition to the pursuit of lower reconstruction error between $\hat{\boldsymbol{x}}_p$ and $\boldsymbol{x}_p$, we also expect that their latent characteristics are equal. Similar to deep metric learning [31], we employ a learnable function f to map $\hat{\boldsymbol{x}}_p$ and $\boldsymbol{x}_p$ into an embedding space and then compare their outputs in that space. Here, we consider a f that is the feature mapping in the intermediate layer of the discriminator. Then, a natural choice is to match $f(\boldsymbol{x}_p)$ with $f(\hat{\boldsymbol{x}}_p)$ by using mean square error $\mathcal{L}_{\mathrm{fm}} = \mathbb{E}[(1/L)\|f(\boldsymbol{x}_p) - f(\hat{\boldsymbol{x}}_p)\|_2^2]$, where L is the length of vector $f(\boldsymbol{x}_p)$, and we call this operation as ***feature matching***. In practice, it enhances the training stability of the discriminator and helps the unmixing model achieve a better solution.

C. Total Loss

As stated before, our objective function consists of three parts: the underlying SAD loss, the Wasserstein distance, and the feature matching. Thus, the total loss of our unmixing model is formulated as follows:

$$\mathcal{L}_T = \mathcal{L}_{\mathrm{sad}} + \lambda_1\mathcal{L}_{\mathrm{wd}} + \lambda_2\mathcal{L}_{\mathrm{fm}} \tag{22}$$

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Algorithm 1 JMnet for Hyperspectral Unmixing

Require: Hyperspectral data set $\boldsymbol{X}$, hyperparameters λ_1 and λ_2. m, the batch size. l_a and l_d, the learning rate of autoencoder and discriminator respectively. Optimizer Adam with default settings.

Require: initial autoencoder parameters $\theta = [\boldsymbol{W}, \boldsymbol{E}]$, initial discriminator parameters $\omega = [\boldsymbol{D}_1, \boldsymbol{D}_2]$.

while θ has not converged **do**
  for $p = 1 \ldots m$ **do**
    Sample data $\boldsymbol{x}_p$ from $\boldsymbol{X}$. Compute $\boldsymbol{m}_p$ by (11) and $\hat{\boldsymbol{x}}_p$ by (15). Compute $f(\boldsymbol{x}_p)$ and $f(\hat{\boldsymbol{x}}_p)$ by (19). L is the length of vector $f(\boldsymbol{x}_p)$.
    $\mathcal{L}_d^{(p)} = -\mathcal{D}_\omega(\boldsymbol{x}_p) + \mathcal{D}_\omega(\hat{\boldsymbol{x}}_p)$
    $\mathcal{L}_a^{(p)} = \mathrm{SAD}(\boldsymbol{x}_p, \hat{\boldsymbol{x}}_p) - \lambda_1 \mathcal{D}_\omega(\hat{\boldsymbol{x}}_p) + \frac{\lambda_2}{L} \left\| f(\boldsymbol{x}_p) - f(\hat{\boldsymbol{x}}_p) \right\|_2^2$
  end for
  $\omega = \mathrm{Adam}(\nabla_\omega \frac{1}{m} \sum_{p=1}^{m} \mathcal{L}_d^{(p)}, \omega, l_d)$
  $\theta = \mathrm{Adam}(\nabla_\theta \frac{1}{m} \sum_{p=1}^{m} \mathcal{L}_a^{(p)}, \theta, l_a)$
end while
Return: $\boldsymbol{E}$ and $\{\boldsymbol{m}_p\}_{p=1}^m$.

where λ_1 and λ_2 are treat as regularization parameters. Since our proposed network architecture includes two modules, an autoencoder and a discriminator, we need to optimize their corresponding parameters ω and θ alternately. The objective functions of the autoencoder and discriminator are represented by $\mathcal{L}_a$ and $\mathcal{L}_d$, respectively. Specifically, $\mathcal{L}_{\mathrm{wd}}$ is an approximation of the Wasserstein distance between $\{\boldsymbol{x}_p\}_{p=1}^m$ and $\{\hat{\boldsymbol{x}}_p\}_{p=1}^m$, which is computed through updating ω. $\mathcal{L}_{\mathrm{wd}}$ is equivalent to minimizing the following form:

$$\mathcal{L}_d = -\mathbb{E}(\mathcal{D}_\omega(\boldsymbol{x}_p)) + \mathbb{E}(\mathcal{D}_\omega(\hat{\boldsymbol{x}}_p)). \tag{23}$$

Note that we aim to minimize the Wasserstein distance between $\{\boldsymbol{x}_p\}_{p=1}^m$ and $\{\hat{\boldsymbol{x}}_p\}_{p=1}^m$, where $\{\hat{\boldsymbol{x}}_p\}_{p=1}^m$ are produced by the autoencoder parameterized with θ. Next, we need to solve $\min_\theta \mathcal{L}_{\mathrm{wd}}$, which is equivalent to minimizing $-\mathbb{E}(\mathcal{D}_\omega(\hat{\boldsymbol{x}}_p))$ when ω is fixed. Besides, our goal aims to $f(\hat{\boldsymbol{x}}_p)$ closing to $f(\boldsymbol{x}_p)$, so both feature matching and $-\mathbb{E}(\mathcal{D}_\omega(\hat{\boldsymbol{x}}_p))$ belong to the objective function of the autoencoder. That is,

$$\mathcal{L}_a = \mathcal{L}_{\mathrm{sad}} - \lambda_1 \mathbb{E}(\mathcal{D}_\omega(\hat{\boldsymbol{x}}_p)) + \lambda_2 \mathbb{E}\left[\frac{1}{L} \left\| f(\boldsymbol{x}_p) - f(\hat{\boldsymbol{x}}_p) \right\|_2^2\right]. \tag{24}$$

To sum up, the entire process of the proposed unmixing model is outlined in Algorithm 1. For ease of understanding our proposed model, we show the flowing path of the model's gradients in Fig. 2. As shown in this figure, the gradients of the autoencoder are from three parts, corresponding to the three terms of $\mathcal{L}_a$. Note that $\mathcal{L}_d$ provides gradient information for the discriminator. It can be seen that, compared to the single SAD loss, our model obtain more useful gradient information so that the autoencoder can achieve a solution with better unmixing performance, and experimental results in Section IV also confirm it.

D. Model Analysis

In this section, we will give an analysis of the two regularizers in our proposed model. According to Salimans *et al.* [37], adding the feature matching for the autoencoder is to ease overtraining of the current discriminator, so as to address the instability of the discriminator. Meanwhile, our empirical results show that the feature matching indeed enhances the training stability of the discriminator and helps our unmixing model achieve a better solution.

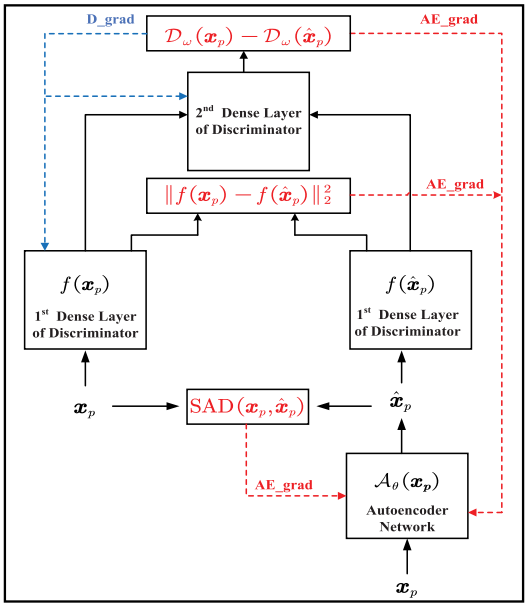
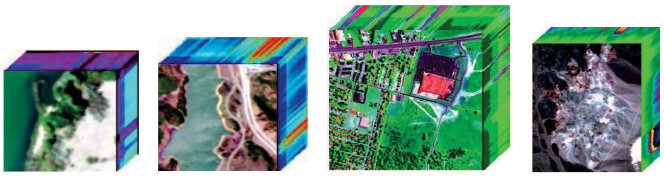


Fig. 2. Flowing path of the gradients. AE_grad represents the gradient flowing to the autoencoder, while D_grad represents the gradient flowing to the discriminator.

Here, we theoretically explain why the Wasserstein distance was chosen as the distribution similarity metric. It will be illustrated from two aspects: continuity and convergence. Remark that the weaker the topology induced by a distance, the easier it for a sequence of distributions to converge. Continuity implies that there is a sequence of parameters θ_t converging to θ; the distributions $\mathbb{P}_{\theta_t}$ also converge to $\mathbb{P}_\theta$ [38].

Continuity: Let $\boldsymbol{Z}$ denote the space formed by all spectral vectors and fix its distribution be $\mathbb{P}$. Any spectral vector $\boldsymbol{z} \in \boldsymbol{Z}$ can be seen as a variable. Let $\mathbb{P}_\theta$ be the distribution of $\mathcal{A}_\theta(\boldsymbol{z})$, where $\mathcal{A}_\theta(\cdot)$ has been defined in (15). Here are three properties: 1) if $\mathcal{A}_\theta(\cdot)$ is continuous in θ, so is $\mathrm{W}(\mathbb{P}, \mathbb{P}_\theta)$; 2) if $\mathcal{A}_\theta(\cdot)$ is locally Lipschitz continuous in θ, then $\mathrm{W}(\mathbb{P}, \mathbb{P}_\theta)$ is continuous everywhere and differentiable almost everywhere;

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(a) (b) (c) (d)

Fig. 3. Cube images of real hyperspectral data sets. (a) Samson. (b) Jasper Ridge. (c) Urban. (d) Cuprite.

and 3) statements 1 and 2 are false for $\mathrm{KL}(\mathbb{P}\|\mathbb{P}_\theta)$ and $\mathrm{JS}(\mathbb{P}\|\mathbb{P}_\theta)$.

Convergence: Let $(\mathbb{P}_n)_{n\in\mathbb{N}}$ be a sequence of distribution on $\boldsymbol{Z}$ and consider all limits as $n \to \infty$. If $\mathrm{KL}(\mathbb{P}\|\mathbb{P}_n) \to 0$ or $\mathrm{KL}(\mathbb{P}_n\|\mathbb{P}) \to 0$, then $\mathrm{JS}(\mathbb{P}_n, \mathbb{P}) \to 0$. $\mathrm{JS}(\mathbb{P}_n, \mathbb{P}) \to 0$ implies the $\mathrm{W}(\mathbb{P}_n, \mathbb{P}) \to 0$. Meanwhile, $\mathbb{P}_n \xrightarrow{\mathcal{D}} \mathbb{P}$ is equivalent to $\mathrm{W}(\mathbb{P}_n, \mathbb{P}) \to 0$, where $\xrightarrow{\mathcal{D}}$ represents convergence in the sense of distribution.

The above two properties indicate that the Wasserstein distance is a much more sensible loss function than KL divergence and JS divergence. It means that the Wasserstein distance can improve the stability of the generative model by providing the gradient information continuously. It should be noted that the function $\mathcal{A}_\theta(\cdot)$ of our model is continuous in θ, which is a composition of a series of continuous functions. Therefore, we can obtain more discriminative features using it as the distribution similarity metric when solving the HSU task. The strict proofs of continuity and convergence have been shown in [30]. The quantitative analysis of these two regularizers can be referred to Section IV-E.

IV. Experiments

In this section, four real hyperspectral data sets are employed to test the performance of our newly developed JMnet. First, a detailed description of data sets is given; next, we summary the baselines and evaluation metrics and follow lists implementation details of each data set. Finally, we conduct a quantitative analysis of the experimental results and analyze the sensitivity of the regularization parameters.

A. Real Hyperspectral Data Sets

This section will present the details of four hyperspectral data sets used in the experiment, as shown in Fig. 3.

Samson is a simple data obtained by the SAMSON sensor, as shown in Fig. 3(a), which has 952×952 pixels with 156 channels covering the wavelengths from 0.4 to 0.9 μm. Since the original image is very huge and in order to make a fair comparison, we consider a subimage of the original data with a size of 95×95. There consists of three constituent materials in the selected scene, including Soil (#1), Tree (#2), and Water (#3).

Jasper Ridge is a common hyperspectral data captured by the AVIRIS sensor that consists of 512×614 pixels. Each pixel has 224 channels ranging from 0.38 to 2.5 μm. Since the original hyperspectral image is too complex to acquire the ground truth, a subimage of 100×100 pixels is considered, as shown in Fig. 3(b). After removing the channels (1–3, 108–112, 154–166, and 220–224) of water vapor absorption and atmospheric influence, there are 198 channels left. There exist four materials in the observed scene: Tree (#1), Water (#2), Soil (#3), and Road (#4).

Urban is a widely researched data set collected by the HYDICE sensor, as shown in Fig. 3(c). Its pixel resolution is 307×307, and there are 210 wavelengths ranging from 0.4 to 2.5 μm. After removing channels 1–4, 76, 87, 101–111, 136–153, and 198–210, we remain 162 channels. Five constituent materials are in the scene: Asphalt (#1), Grass (#2), Tree (#3), Roof (#4), and Dirt(#5).

Cuprite is a challenging data set in the HSU area due to the high correlation among its constituent materials. It is acquired through the AVIRIS over Cuprite, NV, USA, which contains 224 channels with wavelengths covering from 0.37 to 2.48 μm. Its subregion with a size of 250×191 is considered, as shown in Fig. 3(d). The noisy channels (1 and 2 and 221–224) and water absorption channels (104–113 and 148–167) are also removed from data; 12 unique mineral spectral signatures are host in this data set as in [16] and [24]: Alunite (#1), Andradite (#2), Buddingtonite (#3), Dumortierite (#4), Kaolinite1 (#5), Kaolinite2 (#6), Muscovite (#7), Montmorillonite (#8), Nontronite (#9), Pyrope (#10), Sphene (#11), and Chalcedony (#12).

B. Baselines and Metrics

1) Baselines: Our proposed JMnet method will be compared with the following seven unmixing methods.

1) *Vertex Component Analysis (VCA) [12]:* The code can be found in http://www.lx.it.pt/bioucas/code.htm.

2) *Distance-MaxD (DMaxD) [8]:* The code can be downloaded from https://sites.google.com/site/robheylenresearch/code. We apply L_2 kernel as the distance metric for all data sets.

3) *R-CoNMF [14]:* The code can be referred to https://github.com/jubinsoni/Hyperspectral_Unmixing/tree/master/CoNMF_new. We directly use the default settings in this code for all data sets.

4) $L_{1/2}$ *Sparsity-Constrained NMF [13] ($L_{1/2}$-NMF):* The author did not publish the code; we implement it ourselves. We use the result of DmaxD as the endmember initial matrix and initialize the abundance matrix randomly with uniform distribution $U[0, 1]$. The regularization parameter λ is estimated by the sparseness criterion referred to in the literature, and the sum-to-one constraint parameter δ is set to 10.

5) *Endnet [24]:* Since the code is unavailable from the author, we completed it independently. The hyperparameter settings in the objective function are the same as the original literature. We initialize the parameters with uniform distribution $U[-0.1, 0.1]$. For the Urban data set, we use the softmax function to satisfy the sum-to-one constrain and set the batch size into 5. These two settings of the remaining three data sets are consistent with the original literature.

6) *OSPAE [26]:* The code is unavailable from the author, but the author provided us their experimental results, so we referred to their reported scores directly.

7) *DCAE [27]:* Since the code is unavailable from the author, we completed it independently. The literature presents multiple losses and initialization methods; we choose SAD as

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TABLE II

MEAN SAD ($\times 10^{-2}$) AND RMSE ($\times 10^{-2}$) WITH THEIR RESPECTIVE STANDARD DEVIATIONS ON THE SAMSON DATA SET

Endmember(SAD)	VCA	DmaxD	R-CoNMF	$L_{1/2}$-NMF	Endnet	OSPAE	DCAE	JMnet
#1	2.36±0.0	4.04±0.0	2.22±0.2	6.41±2.0	**1.45±0.1**	**0.72±0.0**	1.81±0.8	1.51±0.2
#2	4.90±0.2	**2.19±0.0**	3.63±0.2	4.12±0.7	3.35±0.1	4.37±0.4	3.81±0.4	**3.14±0.1**
#3	48.81±41.23	13.04±0.0	17.78±0.2	26.54±2.8	**2.48±0.3**	5.22±1.9	6.82±1.2	**2.14±0.1**
Average	18.41±14.1	6.42±0.0	7.87±0.2	12.35±1.0	**2.43±0.2**	3.43±0.8	4.12±0.6	**2.26±0.1**
Abundance(RMSE)	VCA	DmaxD	R-CoNMF	$L_{1/2}$-NMF	Endnet	OSPAE	DCAE	JMnet
#1	18.13±3.3	21.68±0.0	19.43±0.1	25.39±3.3	**9.49±0.1**	13.19±1.2	12.68±1.0	**7.63±0.1**
#2	12.63±1.0	12.02±0.0	21.04±0.1	20.41±5.6	**7.24±0.1**	12.27±1.3	8.91±1.3	**5.65±0.1**
#3	19.10±3.7	15.37±0.0	34.26±0.0	24.63±3.7	14.1±0.1	**6.48±0.7**	10.81±1.5	**5.97±0.1**
Average	16.59±0.6	16.36±0.0	24.88±0.0	23.47±4.0	**10.27±0.2**	10.64±1.1	10.80±0.6	**6.42±0.1**

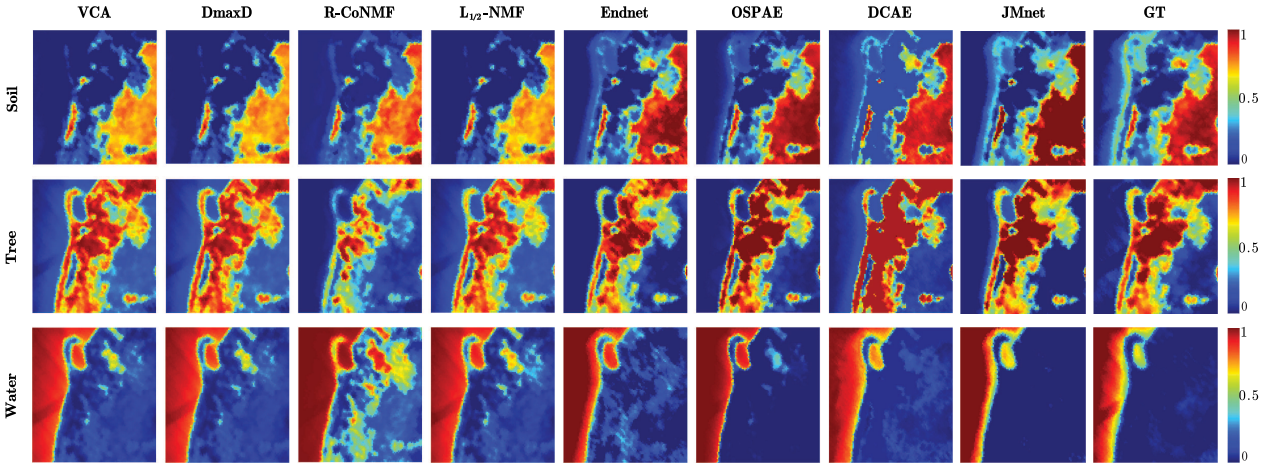


Fig. 4. Visual abundance maps of the Samson data set under different methods and the ground truths.

TABLE III

MEAN SAD ($\times 10^{-2}$) AND RMSE ($\times 10^{-2}$) WITH THEIR RESPECTIVE STANDARD DEVIATIONS ON THE JASPER RIDGE DATA SET

Endmember(SAD)	VCA	DmaxD	R-CoNMF	$L_{1/2}$-NMF	Endnet	OSPAE	DCAE	JMnet
#1	14.78±2.1	15.58±0.0	17.91±1.0	16.09±0.8	6.97±0.4	**3.78±0.8**	10.10±4.1	**2.11±0.1**
#2	29.04±12.1	25.39±0.0	34.63±3.6	36.28±2.0	**3.56±0.9**	**4.63±0.3**	7.28±3.8	6.60±0.2
#3	14.34±3.0	13.35±0.0	18.77±0.6	10.75±1.2	**5.41±0.3**	7.85±3.7	6.33±3.2	**4.33±0.6**
#4	10.69±0.0	10.69±0.0	5.15±0.6	5.33±1.6	**2.61±0.2**	**2.88±0.5**	4.81±2.0	3.52±0.8
Average	17.21±3.7	16.25±0.0	19.11±1.1	17.11±0.3	**4.64±0.5**	4.78±1.9	7.10±0.7	**4.14±0.1**
Abundance(RMSE)	VCA	DmaxD	R-CoNMF	$L_{1/2}$-NMF	Endnet	OSPAE	DCAE	JMnet
#1	12.31±2.3	13.51±0.0	17.84±0.3	17.36±4.0	9.11±0.6	**5.86±1.4**	8.50±1.0	**5.72±0.1**
#2	5.48±1.2	5.22±0.0	21.02±0.6	16.38±0.6	**4.81±0.8**	7.91±1.8	5.78±1.0	**4.39±0.3**
#3	14.56±1.1	14.70±0.0	15.47±0.3	14.11±1.0	11.12±0.3	**7.58±0.6**	14.84±1.1	**7.65±0.5**
#4	12.42±1.5	12.38±0.0	12.63±0.5	11.98±0.5	12.94±0.4	**7.15±0.8**	15.32±0.8	**8.6±0.7**
Avergae	11.20±0.9	11.45±0.0	16.75±0.2	14.85±2.5	9.47±0.6	**7.12±1.1**	11.11±0.5	**6.79±0.3**

the loss function and initialize the parameters by a normal distribution with a standard deviation of 0.1. Settings such as batch size and learning rate are consistent with the original literature.

It should be noted that VCA and DmaxD can only extract the endmembers, and we apply the Hapke model of the MLM method to obtain the abundances [20].

2) Metrics: Two metrics are utilized to evaluate the unmixing performance, namely, SAD and root mean square error (RMSE), which are used to assess the quality of estimated endmember and abundance, respectively

$$\mathrm{SAD}(\boldsymbol{e}_k, \hat{\boldsymbol{e}}_k) = \arccos\left(\frac{\boldsymbol{e}_k \cdot \hat{\boldsymbol{e}}_k}{\|\boldsymbol{e}_k\|_2 \|\hat{\boldsymbol{e}}_k\|_2}\right) \tag{25}$$

$$\mathrm{RMSE}(\boldsymbol{m}_k, \hat{\boldsymbol{m}}_k) = \sqrt{\frac{1}{P}\|\boldsymbol{m}_k - \hat{\boldsymbol{m}}_k\|_2^2} \tag{26}$$

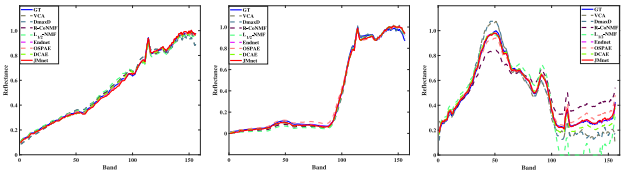
where $\hat{\boldsymbol{e}}_k$ and $\hat{\boldsymbol{m}}_k$ are the kth estimated endmember and abundance, respectively, while $\boldsymbol{e}_k$ and $\boldsymbol{m}_k$ are the kth real

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TABLE IV

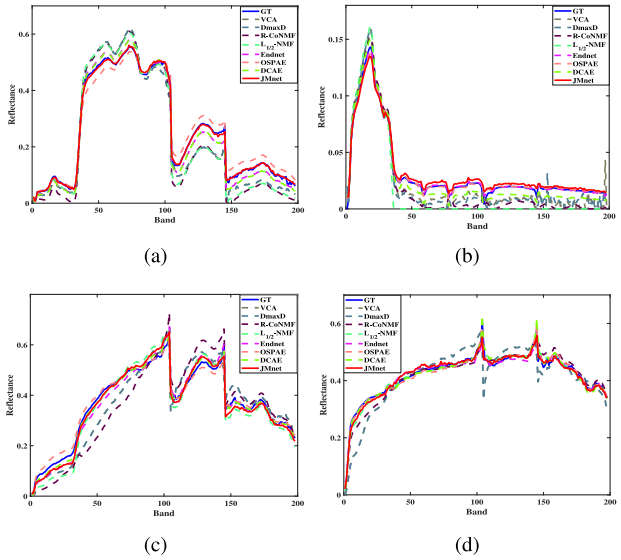
MEAN SAD ($\times 10^{-2}$) AND RMSE ($\times 10^{-2}$) WITH THEIR RESPECTIVE STANDARD DEVIATIONS ON THE URBAN DATA SET

Endmember(SAD)	VCA	DmaxD	R-CoNMF	$L_{1/2}$-NMF	Endnet	OSPAE	DCAE	JMnet
#1	14.80±0.0	14.8.±0.0	38.92±22.6	21.37±5.0	**5.30±0.7**	8.11±1.1	6.17±0.7	**4.30±1.3**
#2	22.27±21.3	65.70±0.0	42.15±11.9	65.11±6.6	**5.11±1.0**	8.68±1.1	6.43±1.8	**5.73±0.6**
#3	11.51±2.7	**6.55±0.0**	13.12±1.7	14.21±1.7	15.05±3.6	**6.15±0.5**	15.11±3.8	12.11±0.6
#4	17.40±0.0	7.98±0.0	7.59±1.7	24.42±8.6	**4.91±1.0**	6.01±1.3	8.80±7.4	**5.66±0.2**
#5	>99.0±9.4	12.15±0.0	10.62±6.7	34.28±7.7	17.20±2.3	**5.17±0.9**	18.37±4.1	**6.61±1.7**
Average	36.55±4.8	21.43±0.0	22.48±3.5	25.93±4.3	9.48±1.6	**6.83±1.7**	10.95±1.8	**6.85±1.7**
Abundance(RMSE)	VCA	DmaxD	R-CoNMF	$L_{1/2}$-NMF	Endnet	OSPAE	DCAE	JMnet
#1	33.40±2.6	36.57±0.0	28.81±7.1	29.40±1.3	18.92±1.1	16.55±2.1	**13.98±1.4**	**10.20±0.6**
#2	41.98±7.3	50.04±0.0	34.12±8.2	48.53±3.1	23.67±2.2	**15.91±1.6**	19.03±2.2	**11.77±1.1**
#3	32.27±8.1	38.35±0.0	24.58±1.5	35.15±0.4	**10.61±1.3**	15.59±2.3	11.84±1.4	**9.10±0.6**
#4	21.25±1.1	22.18±0.0	15.64±1.1	18.98±0.4	11.74±1.2	**10.43±1.1**	12.65±1.3	**7.89±1.1**
#5	30.91±3.8	36.43±0.0	16.10±0.5	17.61±5.0	23.42±1.0	**14.11±1.1**	16.27±3.5	**10.46±0.6**
Average	31.96±3.0	36.72±0.0	23.85±1.2	29.93±1.6	17.67±0.3	**14.52±0.6**	14.76±1.1	**9.88±0.6**



(a) (b) (c)

Fig. 5. Estimated endmembers of different methods on the Samson data set. (a) Soil. (b) Tree. (c) Water.



(a) (b) (c) (d)

Fig. 6. Estimated endmembers of different methods on the Jasper data set. (a) Tree. (b) Water. (c) Soil. (d) Road.

endmember and abundance. Note that the smaller the two metrics, the better the unmixing performance.

C. Implementation Details

The batch size is a critical hyperparameter in neural network optimization. It has been observed, in practice, that a larger batch would lead to degradation of model quality, as measured by its generalization ability [39]. Meanwhile, the smaller the batch size, the fewer epoch when the model converges to minima [40]. Therefore, we empirically choose a relatively small batch size to obtain better performance.

TABLE V

HYPERPARAMETER SETTINGS FOR REAL HYPERSPECTRAL DATA SETS

Hyper-paras / Dataset	λ_1	λ_2	epoch	batch size	STO
Samson	0.01	0.01	100	16	ℓ_1-norm
Jasper Ridge	0.01	0.01	70	16	ℓ_1-norm
Urban	0.01	0.01	20	5	softmax
Cuprite	0.001	0.001	100	4	softmax

Unlike [23], [24] that use VCA-like methods to initialize endmember matrix $\boldsymbol{E}$, we found that initializing all parameters in a random manner would be better. The initializations of the parameters θ and ω subject to the normal distribution with a standard deviation of 0.01 in this article. We conducted experiments and found that, for the Samson and Jasper Ridge data sets, the results obtained by ℓ_1-norm are better than that by softmax, while, for the Urban and Cuprite data sets, the results obtained by softmax are much better than that by ℓ_1-norm. Therefore, we apply ℓ_1-norm for Samson and Jasper Ridge while softmax for Urban and Cuprite. The Adam solver [41] is exploited to update the parameters. Also, we set the learning rate of autoencoder and discriminator to 10^{-4} and 10^{-6}, respectively, for all data sets. Our proposed JMnet method is implemented on Python under the Tensorflow framework. The specific hyperparameter settings of each data set are shown in Table II.

D. Performance Evaluation

In this section, we compare with baselines to test the performance of our method. For a reliable assessment, we repeat each unmixing method on every data set ten times, and the mean and standard deviation of the results are given.

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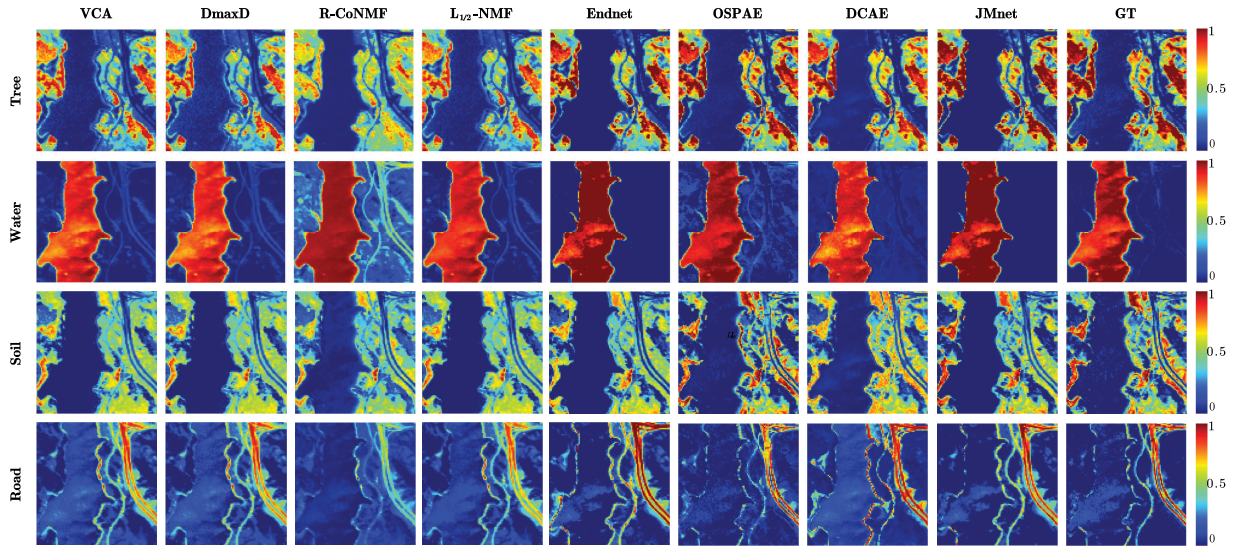


Fig. 7. Visual abundance maps of the Jasper data set under different methods and the ground truths.

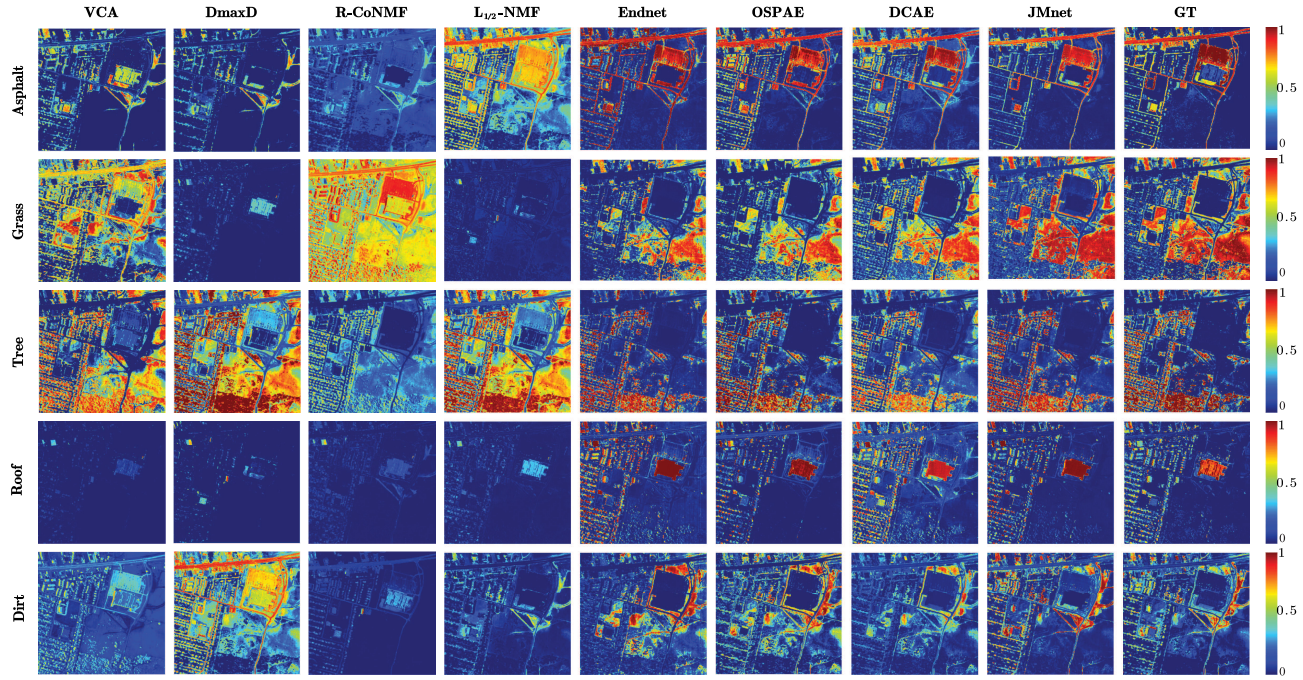


Fig. 8. Visual abundance maps of the Urban data set under different methods and the ground truths.

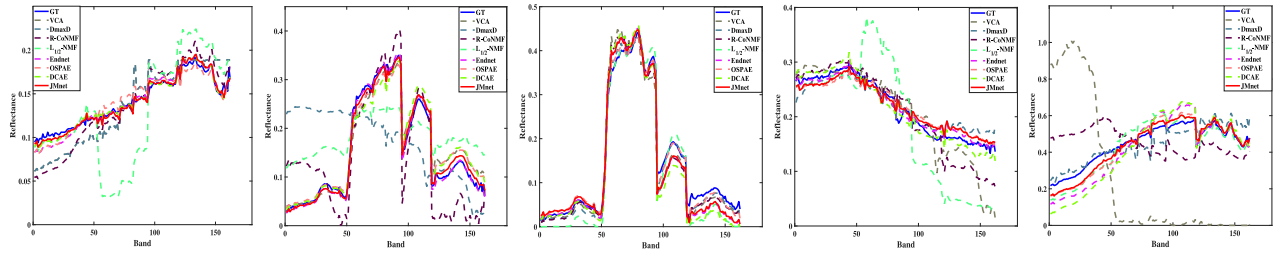


Fig. 9. Estimated endmembers of different methods on the Urban data set. (a) Asphalt. (b) Grass. (c) Tree. (d) Roof. (e) Dirt.

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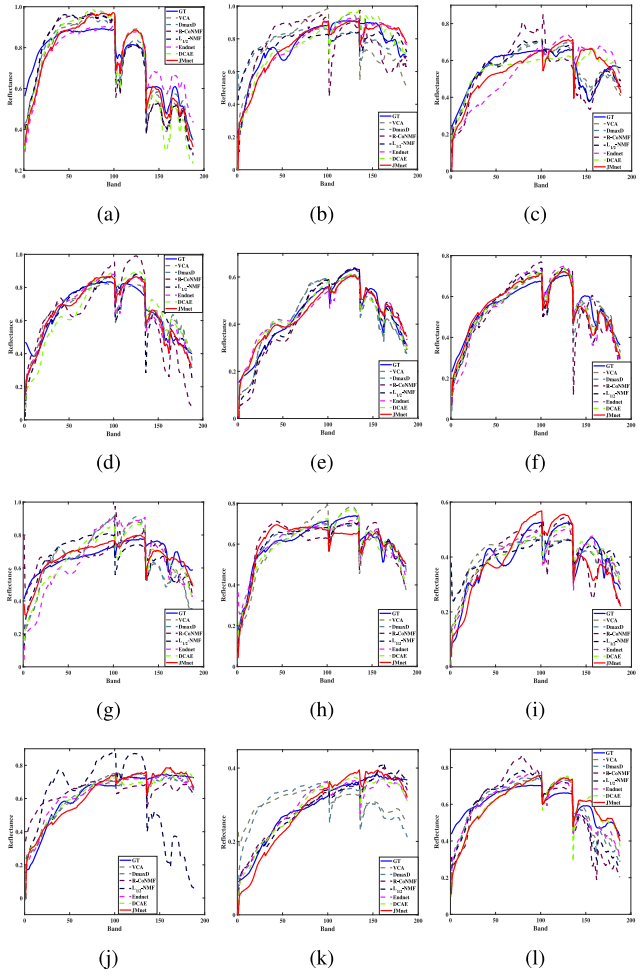


Fig. 10. Estimated endmembers of different methods on the Cuprite data set. (a) Alunite. (b) Andradite. (c) Buddingtonite. (d) Dumortierite. (e) Kaolinite$_1$. (f) Kaolinite$_2$. (g) Muscovite. (h) Montmorillonite. (i) Nontronite. (j) Pyrope. (k) Sphene. (l) Chalcedony.

At the same time, we provide the abundance maps and the endmember signatures that are obtained by different methods and the ground truths (GT).

The quantitative results, including SAD and RMSE, are summarized in Tables III–VI, where the best results are illustrated in red bold, while the second best results are illustrated in blue bold. Table VI only reports the SAD performance since the abundance ground truth of the Cuprite data set is unavailable. Each row of these tables provides the SAD or RMSE results of different methods for a single material, and the last row represents the average result of all materials. Figs. 4–6 visualize the abundance maps of each unmixing method and ground truths on the Samson, Jasper Ridge, and Urban data sets, respectively. Fig. 7 only visualizes the abundance maps of JMnet on the Cuprite data set. Meanwhile, the estimated endmember signatures by different methods under four data sets are plotted in Figs. 8–11.

1) Samson: As shown in Table II, the average SAD of our method is better than the second best one, and the individual SAD is competitive with other unmixing methods. Meanwhile, both the average RMSE and the individual RMSE of our

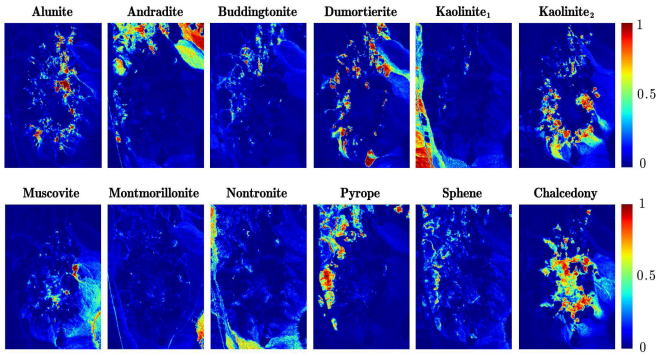


Fig. 11. Estimated abundance maps of the Cuprite data set under the JMnet method.

method achieve the lowest scores. The average RMSE is reduced by 37.5% relating to the second best method. As can be seen from Fig. 5, most unmixing methods can easily extract the correct spectra of endmembers. Fig. 4 provides the visual abundance maps of various methods and ground truths. From Fig. 4, it can be seen that the abundance maps of our method are closer to the ground truths, especially the materials #2 and #3.

2) Jasper Ridge: According to the results in Table III, our method makes a minor improvement on the average SAD of Endnet but reduces the SAD of #1 by 44.2% in comparison with OSPAE. For the abundance comparisons, our method obtained the smallest average RMSE, and the individual RMSE is either the best or the second best. Figs. 6 and 7, respectively, show the estimated endmembers and abundance maps by various methods; it is clear that the four learning-based methods, i.e., JMnet, Endnet, DCAE, and OSPAE, can produce more satisfying results than other traditional methods. From the visual abundance maps in Fig. 7, it can be seen that our results on #1 and #2 are more recognizable, while the results of OSPAE on #3 and #4 are better.

3) Urban: Its experimental results are presented in Table IV, showing that our average SAD is almost the same as OSPAE, while our SAD indicators are slightly inferior to those of OSPAE and Endnet. Specifically, in terms of individual SAD, we get the best result on #1, and second best results on #2, #4, and #5 are still competitive. What stands out in the table is that the average RMSE and individual RMSE of our method are the best. The average RMSE and individual RMSE are reduced by 32.1% and 27% relating to the second best method, respectively. From Fig. 9, we find that endmembers estimated by learning-based methods can be closer to the ground truths. As shown in Fig. 8, our method obtains the best approximation to the ground-truth abundance maps, while traditional methods will appear endmember mixed phenomenon leading to bad results. For example, VCA, DmaxD, R-CoNMF, and $L_{1/2}$-NMF confuse Asphalt, Grass, and Tree. Although learning-based methods, such as Endnet, OSPAE, and DCAE confused nothing, their estimated abundances on #2 and #5 are not ideal, while our results on these two materials are better and look much sharper in visual.

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TABLE VI

MEAN SAD ($\times 10^{-2}$) WITH ITS STANDARD DEVIATION ON THE CUPRITE DATA SET

Endmember	VCA	DmaxD	R-CoNMF	$L_{1/2}$-NMF	Endnet	DCAE	JMnet
#1	**8.73±1.5**	**8.58±0.0**	12.43±1.8	10.66±1.8	16.81±2.5	12.61±3.2	10.20±0.3
#2	10.35±2.0	**7.15±0.0**	13.59±1.2	12.53±0.3	**8.27±0.2**	8.68±4.6	8.49±0.5
#3	**9.71±1.6**	11.58±0.0	13.40±2.2	**6.80±0.2**	13.12±1.1	13.34±1.7	17.6±0.5
#4	10.00±1.7	**8.01±0.0**	15.62±4.3	10.82±0.2	**6.78±0.7**	10.70±3.0	8.34±0.3
#5	9.95±3.4	8.47±0.0	18.13±2.8	**7.42±0.8**	10.17±1.1	**7.52±0.1**	7.86±0.3
#6	**7.31±1.4**	8.52±0.0	11.03±5.4	10.46±1.7	16.33±0.2	7.87±4.5	**6.71±0.2**
#7	13.50±4.0	10.26±0.0	19.46±3.5	31.80±6.1	**8.97±1.1**	16.19±2.4	**8.11±0.1**
#8	**6.10±1.2**	6.19±0.0	11.12±2.6	5.46±0.2	8.53±0.1	**5.28±0.9**	8.22±0.5
#9	9.30±2.9	**7.36±0.0**	12.60±0.6	15.69±0.5	13.67±0.1	11.95±5.4	**9.24±0.1**
#10	12.00±7.5	13.08±0.0	7.18±3.9	9.66±1.7	8.81±0.4	**6.30±0.9**	**7.04±0.2**
#11	21.3±10.5	28.56±0.0	9.36±1.8	27.71±0.4	**5.25±1.2**	**6.65±0.4**	7.79±0.6
#12	**10.90±1.1**	**8.23±0.0**	17.24±5.8	14.56±0.9	13.94±0.9	16.26±3.3	12.60±0.2
Average	10.80±1.0	**10.54±0.0**	13.65±0.6	11.20±0.3	10.89±0.8	10.53±0.8	**9.39±0.1**

TABLE VII

MEAN SAD ($\times 10^{-2}$) RESULTS OF DIFFERENT PARAMETER PAIRS ON THE SAMSON DATA SET

λ_2 \ λ_1	0	0.001	0.01	0.1
0	5.175	2.221	2.265	3.697
0.001	4.743	2.235	2.228	3.693
0.01	2.382	2.169	2.232	3.699
0.1	2.737	2.703	2.422	3.652

TABLE VIII

MEAN RMSE ($\times 10^{-2}$) RESULTS OF DIFFERENT PARAMETER PAIRS ON THE SAMSON DATA SET

λ_2 \ λ_1	0	0.001	0.01	0.1
0	13.92	6.707	6.291	7.024
0.001	13.03	6.837	6.287	7.014
0.01	8.799	7.179	6.284	6.977
0.1	10.26	10.39	7.736	6.927

4) Cuprite: Table VI only shows the SAD performance of different methods. These results illustrate that methods such as VCA and DmaxD based on pure-pixel assumption can obtain good performance. However, our method still gains the best in terms of average SAD. In terms of the individual SAD indicator, four materials get the best score, and one material gets the second best under the DmaxD method, while five materials obtain the second best score under the VCA method. Under our method, two materials get the best score, and two materials get the second best. Endnet and DCAE have the same situation as ours. From Table VI and Fig. 10, we can find that all unmixing methods are competitive except that R-CoNMF is worse. It is apparent from Fig. 11, there exists no overlapping area in the abundance maps, indicating favorable sparsity and satisfactory unmixing results of our method.

E. Regularization Parameter Analysis

There are two regularization parameters in our unmixing model: λ_1 and λ_2. In this section, we will analyze the impact of these two parameters on unmixing performance through conducting experiments on the Samson data set. λ_1 and λ_2 are sampled from $\{0, 0.001, 0.01, 0.1\}$. Tables VII and VIII illustrate the mean SAD and RMSE results under different parameter pairs, respectively. When $\lambda_1 > 0$ and $\lambda_2 = 0$, the mean SAD and RMSE have a sharp decrease compared to the situation when $\lambda_1, \lambda_2 = 0$. We can obtain the similar results on the mean SAD and RMSE when $\lambda_1 = 0$ and $\lambda_2 > 0$. Moreover, sampling λ_1 from $\{0.001, 0.01\}$ and λ_2 from $\{0, 0.001, 0.01\}$ can achieve more stable and better results. The regularization parameters of the other three data sets will also be selected from this region.

Therefore, the above results confirm that combining the proposed two regularization terms with the original SAD loss, indeed, improves the unmixing performance a lot. This supports our view that taking the Wasserstein distance and feature matching as regularizers can promote the unmixing model to achieve a solution with better unmixing performance.

V. CONCLUSION

In this article, we proposed a joint metric neural network for hyperspectral unmixing by introducing the Wasserstein distance and the feature matching as regularization terms. Specifically, a well-designed autoencoder network is used for endmember extraction and abundance estimation with the SAD loss function. Regarding the observation and the reconstruction as two distributions, the Wasserstein distance between them is computed implicitly through a discriminator, while the feature matching is adopted in the intermediate layer of the discriminator to compare the feature difference between them. We theoretically and practically prove that, by combining SAD with the two mentioned regularizers as the final objective, the unmixing model can achieve a more refined solution. Experimental results on four real hyperspectral data sets indicate the effectiveness and superiority of our method compared to baselines.

Our method is based on a shallow dense neural network with the LMM, we can focus on the deep convolutional network in the next work, and theoretical research can be conducted to the feature matching. Furthermore, how to simulate the NLMM through a network might be a hot topic in the future. Finally, injecting the prior information, such as the graph into the network, might achieve better results.

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